Introductory Tutorial

Coherent and Dissipative Transport on Small Quantum Graphs

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April 2026

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1 How to use this tutorial

This document is written for a reader who already knows undergraduate quantum mechanics, statistical physics, and thermodynamics, but has not yet studied in detail open quantum systems, the Lindblad equation, transport on quantum networks, or the meanings of *sink*, *trap*, and *loss*. The goal is not to sound advanced. The goal is to build a clean bridge between concepts you already know and the language used in the simulation laboratory.

2 The physical question in plain language

The central question is:

> How does a quantum excitation move on a small graph when both coherent dynamics and environmental dissipation are present?

In practical terms:

- the graph represents a network of sites;
- the excitation can occupy one of those sites;
- couplings between sites allow the excitation to move;
- the environment can destroy coherence or remove population;
- a special absorbing channel, called a *sink*, represents successful arrival at the target.

This question is directly connected to the literature on environment-assisted transport in small quantum networks [1–4].

3 Graphs, nodes, and edges

Mathematically, a graph is a set of vertices connected by edges. Physically, in this project each **node** is a site that can host the excitation, and each **edge** is a coupling channel between two sites. The main topologies in the current laboratory are:

- **chain**: each node talks only to its nearest neighbors;
- **ring**: a closed chain;
- **complete graph**: every node talks to every other node.

4 Basis states: what does "the excitation is on node i" mean?

If the system has N graph nodes, we use the basis

$$\{|0\rangle, |1\rangle, |2\rangle, \ldots, |N-1\rangle\}. \tag{1}$$

The interpretation is direct:

$$|i\rangle = \text{“the excitation is localized on node } i\text{”}. \tag{2}$$

For a two-site system,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3}$$

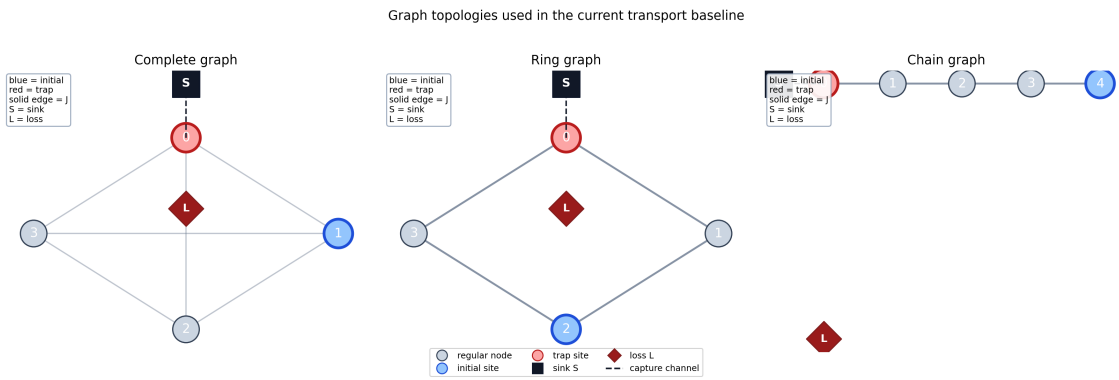

Figure 1: Current baseline topologies. The blue node is the initial site, the red node is the trap site, the square S is the sink, and the square L is the loss channel.

5 Step 1: the simplest coherent walk

We begin with the simplest possible case: two nodes, no sink, no loss, and no dephasing.

5.1 Two-site Hamiltonian

If the two sites are coupled with strength J, the Hamiltonian is

$$H = J(|0\rangle\langle 1| + |1\rangle\langle 0|) = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}. \tag{4}$$

5.2 Opening the unitary-evolution calculation

The unitary evolution operator is

$$U(t) = e^{-iHt}. \tag{5}$$

Since

$$H^2 = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}^2 = J^2\mathbb{I}, \tag{6}$$

the exponential series can be summed:

$$U(t) = \mathbb{I} - iHt - \frac{H^2t^2}{2!} + \frac{iH^3t^3}{3!} + \dots \tag{7}$$

$$= \cos(Jt)\,\mathbb{I} - i\sin(Jt)\,\frac{H}{J}. \tag{8}$$

If the initial state is $|\psi(0)\rangle = |1\rangle$, then

$$|\psi(t)\rangle = U(t)\,|1\rangle \tag{9}$$

$$= \cos(Jt)\,|1\rangle - i\sin(Jt)\,|0\rangle\,. \tag{10}$$

Therefore the node populations are

$$P_0(t) = \sin^2(Jt), \qquad P_1(t) = \cos^2(Jt). \tag{11}$$

This is the simplest coherent walk: the excitation oscillates between the two sites.

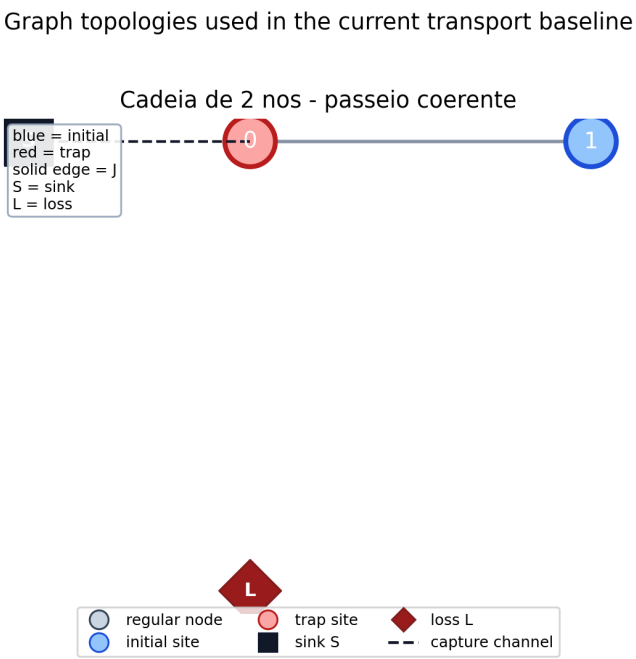

Figure 2: Learning step 1: a two-site chain without sink, loss, or dephasing. The purpose is simply to see the clean coherent oscillation.

6 Why do we move from state vectors to density matrices?

Once the environment can destroy coherence or remove population, the natural object becomes the density matrix:

$$\rho = |\psi\rangle\langle\psi| \tag{12}$$

for pure states, and more generally a statistical mixture for mixed states.

For two sites,

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}. \tag{13}$$

Interpretation:

- ρ_{00} and ρ_{11} are populations;
- ρ_{01} and ρ_{10} are coherences.

7 From the Schrödinger equation to the von Neumann equation

If the system is closed, the state vector obeys

$$i\frac{d}{dt}|\psi\rangle = H|\psi\rangle. \tag{14}$$

Starting from $\rho = |\psi\rangle\langle\psi|$, we derive

$$\begin{aligned} \frac{d\rho}{dt} &= \frac{d}{dt}(|\psi\rangle\langle\psi|) && (15)\\ &= \left(\frac{d|\psi\rangle}{dt}\right)\langle\psi| + |\psi\rangle\left(\frac{d\langle\psi|}{dt}\right) && (16)\\ &= -iH\rho + i\rho H && (17)\\ &= -i[H,\rho]. && (18) \end{aligned}$$

8 Opening the commutator explicitly

Take

$$H = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}, \qquad \rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}. \tag{19}$$

Then

$$[H, \rho] = \begin{pmatrix} J(\rho_{10} - \rho_{01}) & J(\rho_{11} - \rho_{00}) \\ J(\rho_{00} - \rho_{11}) & J(\rho_{01} - \rho_{10}) \end{pmatrix}. \tag{20}$$

Thus

$$\dot{\rho}_{00} = -iJ(\rho_{10} - \rho_{01}), \tag{21}$$
$$\dot{\rho}_{11} = -iJ(\rho_{01} - \rho_{10}), \tag{22}$$
$$\dot{\rho}_{01} = -iJ(\rho_{11} - \rho_{00}), \tag{23}$$
$$\dot{\rho}_{10} = -iJ(\rho_{00} - \rho_{11}). \tag{24}$$

These equations show explicitly that populations and coherences are coupled.

9 Open quantum systems and the Lindblad equation

Now we introduce the main new concept: the system is not perfectly isolated. A standard effective model for such dynamics is the Lindblad master equation:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right). \tag{25}$$

Interpretation:

- $-i[H, \rho]$ is the coherent part;
- L_k are dissipative channels;
- each L_k represents one effective physical process.

This is the standard effective starting point for open quantum systems [5, 6].

10 Dephasing: the first dissipative channel to understand

In the current laboratory, the most important dissipative channel is dephasing. On node i, it is written as

$$L_i^{(\phi)} = \sqrt{\gamma_{\phi,i}}\, |i\rangle \langle i|. \tag{26}$$

For a two-site system with the same dephasing rate γ_ϕ on each site, the dissipative contribution yields

$$\dot{\rho}_{01}\Big|_\phi = -\gamma_\phi\, \rho_{01}, \qquad \dot{\rho}_{10}\Big|_\phi = -\gamma_\phi\, \rho_{10}. \tag{27}$$

So dephasing mainly kills coherence. That is why a modest amount can suppress harmful interference, while too much dephasing can suppress transport itself.

11 What is the sink?

The *sink* is a success channel. It is not an ordinary graph node. It represents

> "the excitation has arrived at the useful target".

If the trap site is $|t\rangle$, then we write

$$L^{(\text{sink})} = \sqrt{\kappa}\, |s\rangle \langle t| \,, \tag{28}$$

where $|s\rangle$ is the sink state.

The main laboratory observable is

$$\eta(T) = \rho_{ss}(T). \tag{29}$$

The sink population therefore measures success, not generic motion through the graph.

12 What is loss?

The *loss* channel is another absorbing state, but it represents failure:

$$L_i^{(\text{loss})} = \sqrt{\Gamma_i}\, |\ell\rangle \langle i| \,. \tag{30}$$

Population leaving the graph can therefore have two destinations:

- **sink**: successful capture;
- **loss**: parasitic dissipation.

13 Laboratory observables

The three main observables are:

1. **sink efficiency**
$$\eta(T) = \rho_{ss}(T); \tag{31}$$
2. **population dynamics**
$$P_i(t) = \rho_{ii}(t); \tag{32}$$
3. **coherence**
$$C_{\ell_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \tag{33}$$

14 Three simple learning simulations

14.1 Step 1: coherent walk

Configuration file:

```
configs/transport_learning_step1_walk.json
```

Here the sink and loss are absent, so $\eta(T) = 0$ by construction. The point is simply to see coherent oscillation.

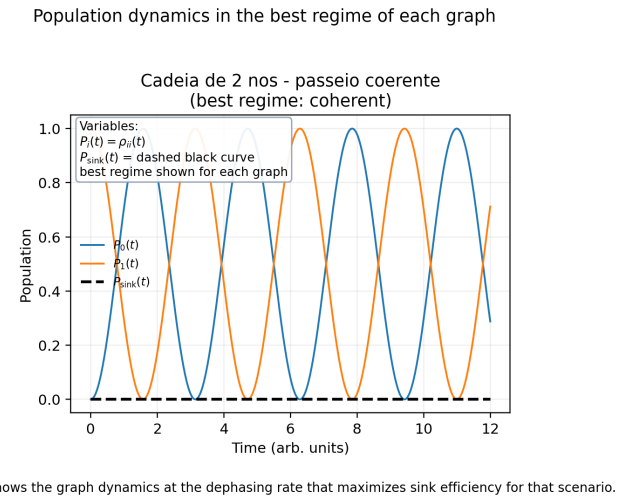

Figure 3: Step 1: population dynamics for a two-site system without dissipation or sink.

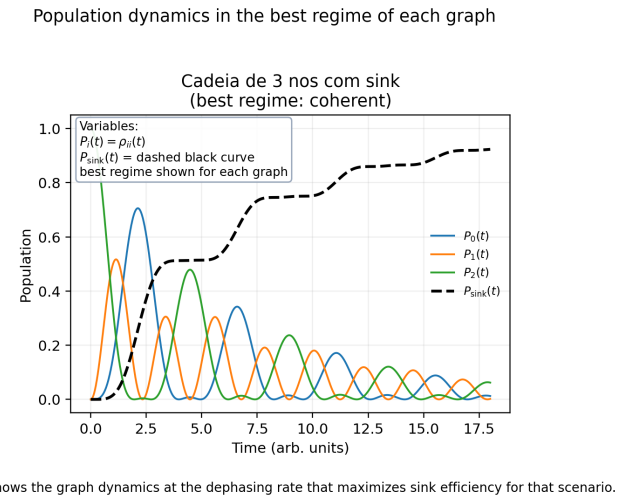

Figure 4: Step 2: with an active sink, the sink population grows in time. This is the first figure that should be read as successful transport.

14.2 Step 2: introducing the sink

Configuration file:

```
configs/transport_learning_step2_sink.json
```

Now the excitation moves through a three-site chain and can be captured once it reaches the trap site.

14.3 Step 3: introducing dephasing

Configuration file:

```
configs/transport_learning_step3_dephasing.json
```

Now the genuinely interesting question appears: can a small amount of dephasing improve transport? In the current example, yes. The maximum efficiency does not occur at $\gamma_\phi = 0$, but at an intermediate value.

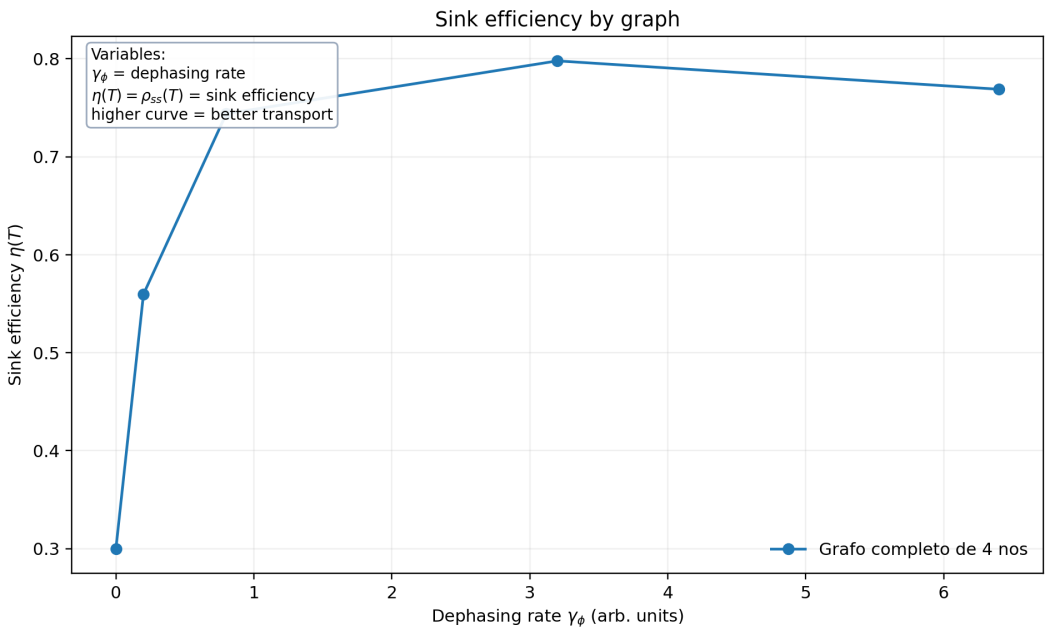

Figure 5: Step 3: an example in which the transport efficiency improves when dephasing increases from zero to an intermediate regime.

Table 1: Learning-path results generated directly from the current simulation artifacts.

Step	Scenario	Best regime	η_{coh}	η_{best}	$\gamma_\phi^{\text{best}}$
Step 1	Two-site coherent walk	coherent	0.000	0.000	0.000
Step 2	Three-site chain with sink	coherent	0.923	0.923	0.000
Step 3	Four-site complete graph with dephasing scan	intermediate	0.300	0.798	3.200

15 What has already been simulated

The current laboratory already contains three pedagogical simulations and one larger baseline comparison.

The present learning-path data say the following:

- **Step 1**: the two-site walk is purely coherent, so the sink efficiency is zero by construction.
- **Step 2**: once the sink is introduced, the three-site chain reaches $\eta \approx 0.923$, so most of the excitation ends in the useful target.
- **Step 3**: on the four-site complete graph, the coherent regime is poor ($\eta \approx 0.300$), but an intermediate dephasing rate improves the efficiency to $\eta \approx 0.798$.

For the larger baseline comparison:

- **Complete graph**: best regime is intermediate, with a strong dephasing-assisted gain.
- **Ring graph**: best regime is coherent for the current parameter set.
- **Chain graph**: best regime is also coherent, and large dephasing is strongly harmful.

Table 2: Baseline graph-comparison results generated from the current laboratory run.

Scenario	Best regime	η_{coh}	η_{best}	$\gamma_\phi^{\text{best}}$
Complete graph	intermediate	0.300	0.798	3.200
Ring graph	coherent	0.795	0.795	0.000
Chain graph	coherent	0.812	0.812	0.000

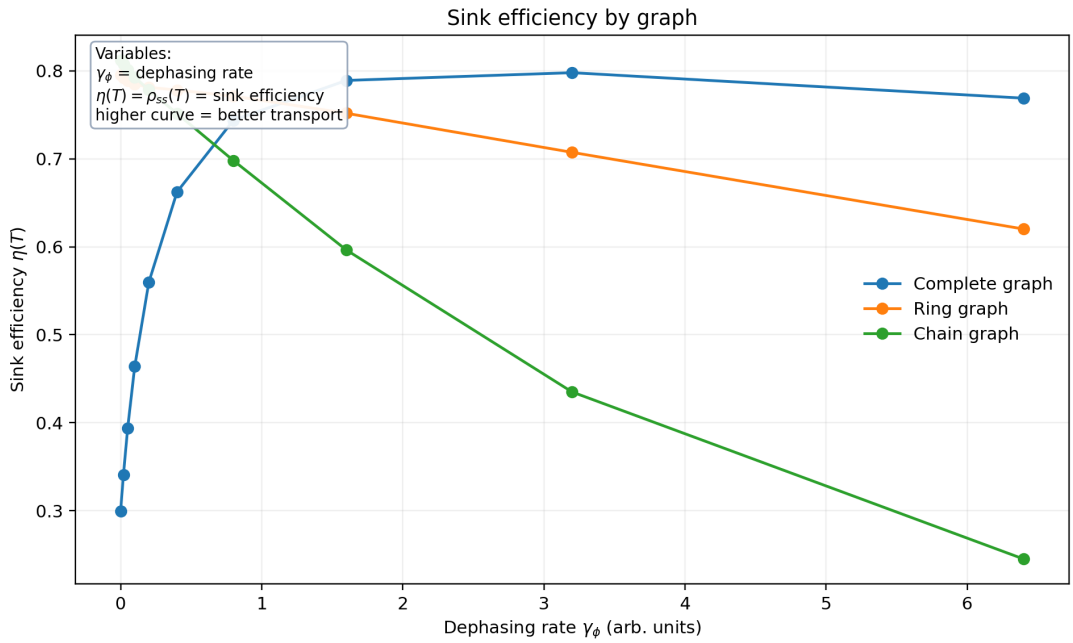

Figure 6: Main baseline comparison between graph topologies. The complete graph exhibits an optimal intermediate regime, whereas the chain and ring are best in the coherent limit for the current parameters.

16 Short solved exercises

Exercise 1: two-site coherent walk

Problem. Consider

$$H = J(|0\rangle\langle 1| + |1\rangle\langle 0|) \tag{34}$$

and the initial state $|\psi(0)\rangle = |1\rangle$. Find $P_0(t)$.

Solution. From the derivation above,

$$|\psi(t)\rangle = \cos(Jt)\,|1\rangle - i\sin(Jt)\,|0\rangle\,, \tag{35}$$

so

$$P_0(t) = |\langle 0|\psi(t)\rangle|^2 = \sin^2(Jt). \tag{36}$$

Exercise 2: sink versus loss

Problem. At the final time T, suppose the state has $\rho_{ss}(T) = 0.80$ and $\rho_{\ell\ell}(T) = 0.15$. What do these numbers mean?

Solution. The sink population is the transport efficiency:

$$\eta(T) = \rho_{ss}(T) = 0.80. \tag{37}$$

So 80% of the excitation ended in the useful target. The loss population says that 15% of the excitation was removed by parasitic dissipation. The remaining population is still distributed among graph nodes.

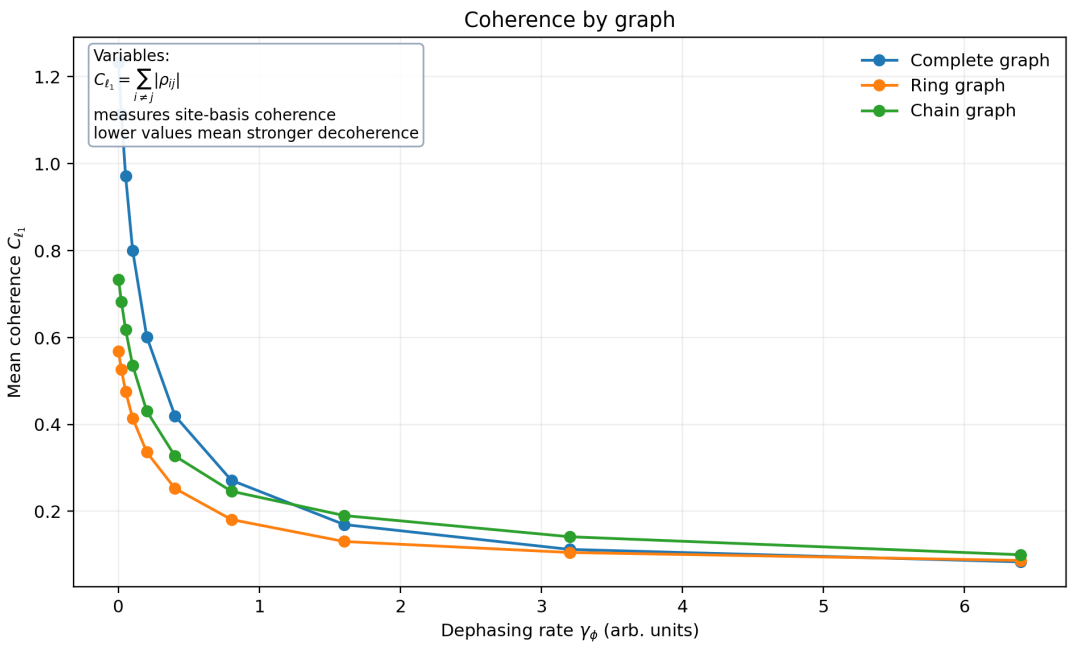

Figure 7: Baseline coherence comparison. The chain preserves the largest coherence in the optimal regime, while the complete graph reaches its best transport efficiency with substantially lower coherence.

Exercise 3: identifying the best regime

Problem. A graph has $\eta_{\text{coh}} = 0.30$, $\eta_{\text{best}} = 0.80$, and $\gamma_\phi^{\text{best}} = 3.2$. Which regime is preferred?

Solution. Since the best efficiency is much larger than the coherent one and it occurs at a nonzero dephasing rate, the preferred regime is **intermediate dephasing**. This is exactly what the current complete-graph simulation shows.

Exercise 4: reading a coherence plot

Problem. If the coherence decreases strongly while the sink efficiency increases, is that automatically a contradiction?

Solution. No. In these transport problems, some coherence is useful, but excessive coherent interference can also trap the excitation in unfavorable pathways. A modest reduction in coherence can therefore improve transport if it suppresses destructive interference without freezing the dynamics.

17 How to use the laboratory without getting lost

The most efficient workflow is:

1. run step 1 and understand coherent oscillation only;
2. run step 2 and understand what the sink actually measures;
3. run step 3 and understand what dephasing changes;
4. only then move to the full baseline comparison.

The most important files are:

- `configs/transport_learning_step1_walk.json`;
- `configs/transport_learning_step2_sink.json`;

- `configs/transport_learning_step3_dephasing.json`;
- `configs/transport_graph_lab_config.json`;
- `scripts/run_transport_learning_path.py`;
- `scripts/run_transport_graph_lab.py`.

18 Closing remarks

If you keep only one idea from this tutorial, let it be this:

> The laboratory is not simulating an arbitrary random walk on a graph. It is simulating how a quantum excitation moves, loses coherence, gets removed by parasitic channels, or reaches the useful target.

In other words:

- the nodes say where the excitation can be;
- the Hamiltonian says how it moves;
- Lindblad terms say how the environment acts;
- the sink measures success;
- the loss channel measures failure;
- the topology changes all of that.

References

[1] Masoud Mohseni, Patrick Rebentrost, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum walks in photosynthetic energy transfer. *The Journal of Chemical Physics*, 129(17):174106, 2008. doi: 10.1063/1.3002335.

[2] Martin B. Plenio and Susana F. Huelga. Dephasing-assisted transport: Quantum networks and biomolecules. *New Journal of Physics*, 10:113019, 2008. doi: 10.1088/1367-2630/10/11/113019.

[3] Patrick Rebentrost, Masoud Mohseni, Ivan Kassal, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum transport. *New Journal of Physics*, 11:033003, 2009. doi: 10.1088/1367-2630/11/3/033003.

[4] Filippo Caruso, Alex W. Chin, Animesh Datta, Susana F. Huelga, and Martin B. Plenio. Highly efficient energy excitation transfer in light-harvesting complexes: The fundamental role of noise-assisted transport. *The Journal of Chemical Physics*, 131(10):105106, 2009. doi: 10.1063/1.3223548.

[5] Heinz-Peter Breuer and Francesco Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, Oxford, 2007. doi: 10.1093/acprof:oso/9780199213900.001.0001.

[6] Ángel Rivas and Susana F. Huelga. *Open Quantum Systems: An Introduction*. Springer, Heidelberg, 2012. doi: 10.1007/978-3-642-23354-8.